

RETAIL SHELF MONITORING USING COMPUTER VISION AND DATA SCIENCE

1. Introduction

Retail stores contain a large number of products displayed on shelves, and maintaining proper product availability and arrangement is an important task. Manual shelf monitoring requires considerable time and may result in human errors. The proposed project, Retail Shelf Monitoring, aims to develop a computer vision-based system that can automatically analyze images of retail shelves. The system will detect products, count them, identify empty or low-stock areas, and detect products that are incorrectly placed.

2. Problem Statement

In retail stores, employees generally monitor shelves manually to check product availability and arrangement. This process is time-consuming and may not identify empty shelves or misplaced products quickly. Therefore, there is a need for an automated system that can analyze shelf images and provide information about product availability and shelf conditions.

3. Objectives

1. To detect products displayed on retail shelves using computer vision.
2. To count the number of products visible on the shelf.
3. To identify empty or low-stock shelf areas.
4. To detect misplaced or incorrectly positioned products.
5. To provide useful information to store employees for better shelf management.
6. To develop a simple dashboard for displaying monitoring results.

4. Proposed System

The proposed system will take an image of a retail shelf as input. Image processing and object detection techniques will be applied to identify products in the image. The system will analyze the detected products and determine whether the shelf has sufficient stock. If an empty or low-stock area is detected, the system can display an alert. The results can be presented through a simple dashboard.

5. Methodology

Step 1 – Data Collection: Collect images of retail shelves containing different products and shelf conditions.

Step 2 – Data Preprocessing: Resize images and perform preprocessing such as normalization and image enhancement where required.

Step 3 – Product Detection: Use an object detection model such as YOLO to identify products in shelf images.

Step 4 – Product Counting: Count detected products and determine the number of visible products for each category.

Step 5 – Shelf Analysis: Analyze the shelf to identify empty spaces, low-stock areas, misplaced products, and properly stocked areas.

Step 6 – Result Visualization: Display detected products and shelf status through a dashboard or visual report.

6. Technologies Required

Programming Language: Python

Computer Vision: OpenCV

Machine Learning/Deep Learning: YOLO / Object Detection

Data Processing: Pandas, NumPy

Visualization: Matplotlib

Dashboard: Streamlit

Development Environment: Jupyter Notebook / Google Colab / VS Code

7. Expected Output

The system is expected to detect products from shelf images, display the number of detected products, identify empty or low-stock areas, highlight incorrectly placed products, provide an overall shelf status, and display results through a simple and user-friendly dashboard.

8. Advantages

- Reduces manual shelf inspection.
- Saves time for retail employees.
- Helps identify stock problems quickly.
- Improves shelf organization.
- Supports better inventory management.
- Can improve product availability for customers.

9. Future Scope

The system can be extended to real-time monitoring using CCTV cameras, automatic inventory updates, notifications to store employees, product price and label detection, sales prediction based on shelf data, and integration with inventory management systems.

10. Conclusion

Retail Shelf Monitoring Using Computer Vision and Data Science is an automated solution for analyzing retail shelf images and identifying product availability and arrangement. By combining computer vision, object detection, and data analysis, the system can reduce manual monitoring and provide useful information to retailers. The project demonstrates a practical application of Data Science and Artificial Intelligence in the retail industry.